INCH-POUND

MIL-C-83723/94D

24 April 1992

SUPERSEDING
MIL-C-83723/94C
31 December 1986

# MILITARY SPECIFICATION SHEET

## CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SINGLE HOLE MOUNT, BAYONET COUPLING, STRAIGHT PIN CONTACT), (SERIES III, CLASSES H AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

(D) The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of specifications and Standards (DODISS) specified in the solicitation: MIL-C-83723.

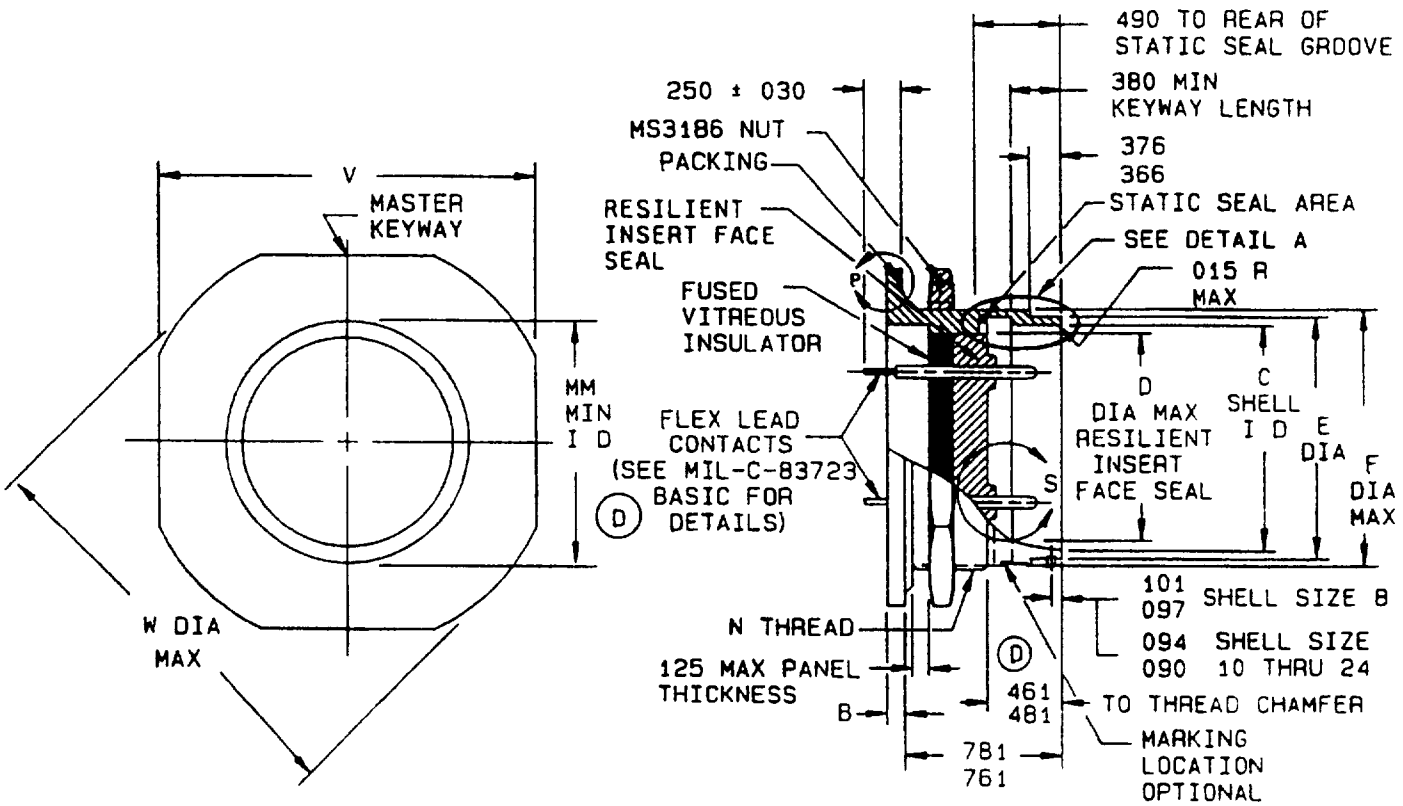



FIGURE 1. Dimensions and configurations.

(D) denotes changes



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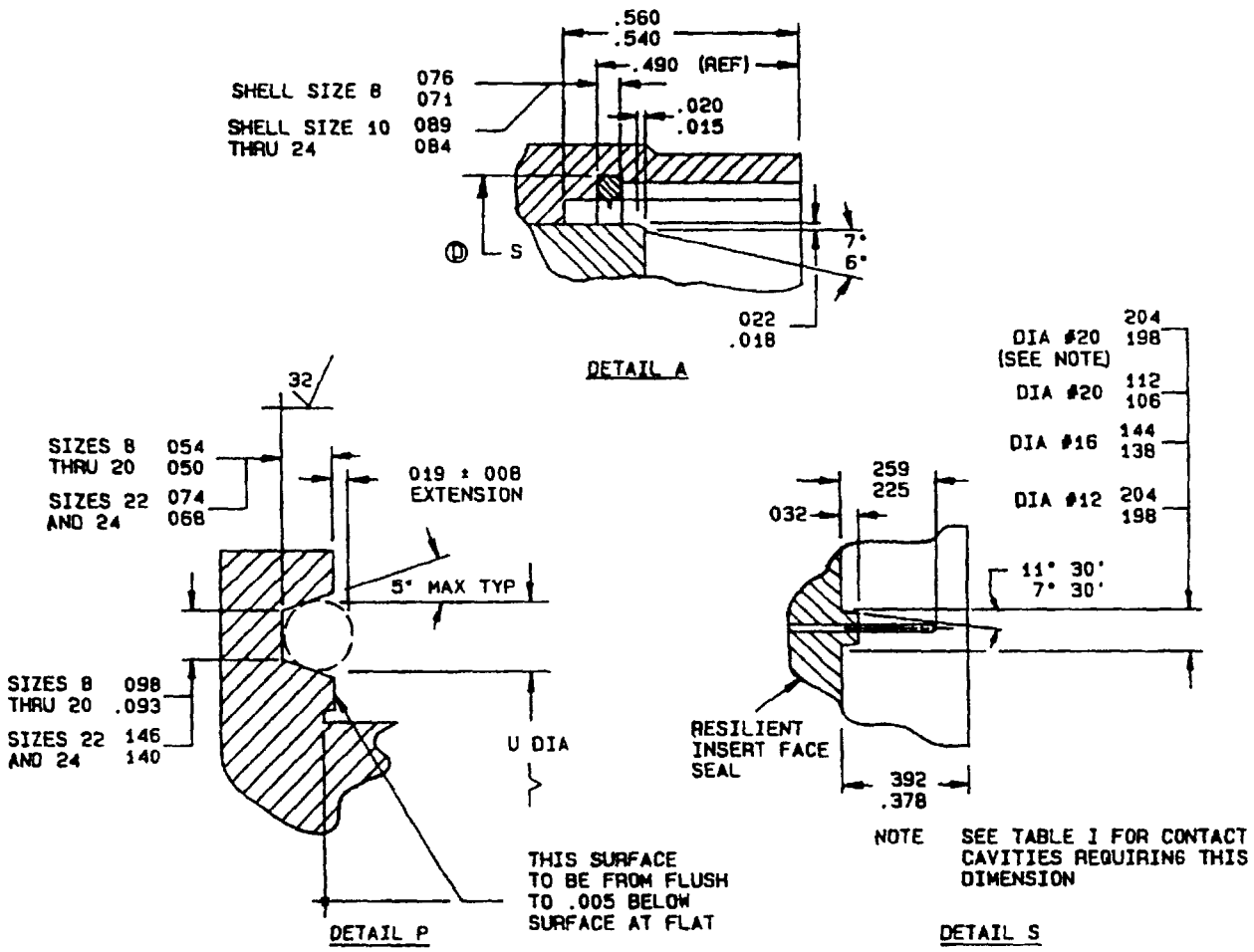



FIGURE 1. Dimensions and configurations - Continued.



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Connectors dimensions.

| Shell size | B | C | D | E | F | N thread | (D) S +.005 -.000 | U |
|---|---|---|---|---|---|---|---|---|
| 8 | .137<br>.097 | .428<br>.433 | .312 | .536<br>.531 | .561 | .625-20<br>UN-2A | .508 | .670 |
| 10 | .137<br>.097 | .530<br>.535 | .388 | .659<br>.654 | .696 | .750-20<br>UNEF-2A | .636 | .777 |
| 12 | .137<br>.097 | .700<br>.705 | .558 | .829<br>.824 | .875 | .9375-20<br>UNEF-2A | .806 | .965 |
| 14 | .137<br>.097 | .769<br>.774 | .627 | .898<br>.893 | .935 | 1.000-20<br>UNEF-2A | .875 | 1.090 |
| 16 | .137<br>.097 | .896<br>.901 | .772 | 1.025<br>1.020 | 1.062 | 1.125-20<br>UN-2A | 1.002 | 1.215 |
| 18 | .137<br>.097 | 1.002<br>1.007 | .860 | 1.131<br>1.126 | 1.187 | 1.250-18<br>UNEF-2A | 1.108 | 1.340 |
| 20 | .137<br>.097 | 1.127<br>1.132 | .985 | 1.256<br>1.251 | 1.312 | 1.375-18<br>UNEF-2A | 1.233 | 1.465 |
| 22 | .168<br>.128 | 1.252<br>1.257 | 1.110 | 1.381<br>1.376 | 1.437 | 1.500-18<br>UNEF-2A | 1.358 | 1.574 |
| 24 | .168<br>.128 | 1.377<br>1.382 | 1.235 | 1.506<br>1.501 | 1.562 | 1.625-18<br>UNEF-2A | 1.483 | 1.699 |

FIGURE 1. Dimensions and configurations - Continued.



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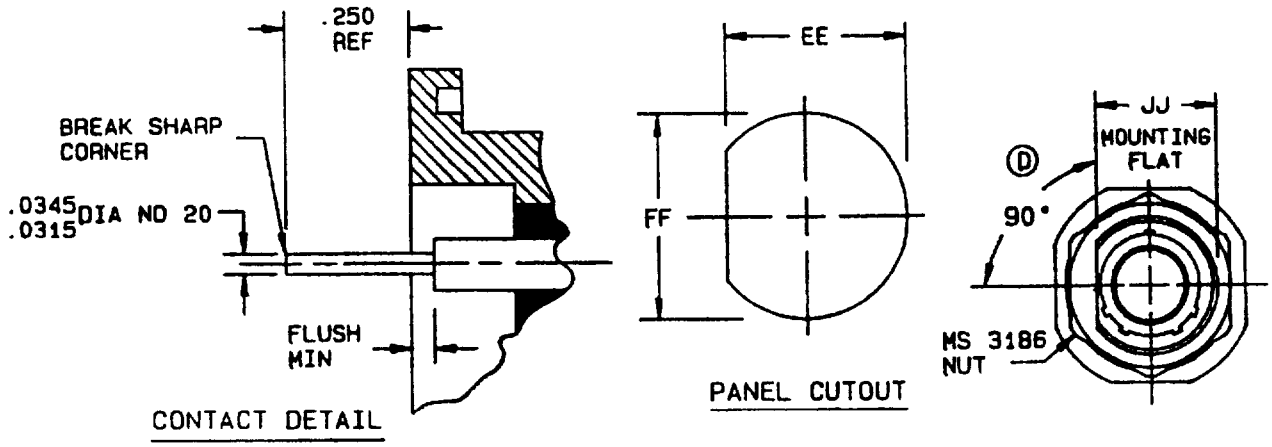
Connectors dimensions - Continued.

| Shell size | V | W | EE | FF | MM | JJ |
|---|---|---|---|---|---|---|
| 8 | .984<br>.974 | 1.068 | .605 | .635 | .493 | .596<br>.590 |
| 10 | 1.109<br>1.099 | 1.192 | .730 | .760 | .555 | .721<br>.715 |
| 12 | 1.296<br>1.286 | 1.380 | .917 | .947 | .743 | .908<br>.902 |
| 14 | 1.396<br>1.386 | 1.505 | .980 | 1.010 | .805 | .971<br>.965 |
| 16 | 1.521<br>1.511 | 1.630 | 1.105 | 1.135 | .931 | 1.096<br>1.090 |
| 18 | 1.646<br>1.636 | 1.756 | 1.229 | 1.260 | 1.055 | 1.220<br>1.214 |
| 20 | 1.771<br>1.761 | 1.860 | 1.354 | 1.385 | 1.243 | 1.345<br>1.339 |
| 22 | 1.959<br>1.949 | 2.068 | 1.479 | 1.510 | 1.305 | 1.470<br>1.464 |
| 24 | 2.084<br>2.074 | 2.160 | 1.604 | 1.635 | 1.493<br>1.452 | 1.595<br>1.589 |

FIGURE 1. Dimensions and configurations - Continued.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| .000050 | 0.0013 | .089 | 2.26 | .259 | 6.58 | .558 | 14.17 | .558 | 14.17 |
| .003 | 0.08 | .090 | 2.29 | .312 | 7.92 | .560 | 14.22 | .560 | 14.22 |
| .004 | 0.10 | .093 | 2.36 | .366 | 9.30 | .561 | 14.25 | .561 | 14.25 |
| .005 | 0.13 | .094 | 2.39 | .376 | 9.55 | .590 | 14.99 | .590 | 14.99 |
| .008 | 0.20 | .097 | 2.46 | .378 | 9.60 | .596 | 15.14 | .596 | 15.14 |
| .010 | 0.25 | .098 | 2.49 | .380 | 9.65 | .605 | 15.37 | .605 | 15.37 |
| .015 | 0.38 | .101 | 2.57 | .388 | 9.68 | .627 | 15.93 | .627 | 15.93 |
| .018 | 0.46 | .106 | 2.69 | .392 | 9.96 | .635 | 16.13 | .635 | 16.13 |
| .019 | 0.48 | .112 | 2.48 | .428 | 10.87 | .636 | 16.15 | .636 | 16.15 |
| .020 | 0.51 | .128 | 3.25 | .433 | 11.00 | .654 | 16.61 | .654 | 16.61 |
| .022 | 0.56 | .137 | 3.48 | .461 | 11.71 | .659 | 16.74 | .659 | 16.74 |
| .030 | 0.76 | .138 | 3.51 | .481 | 12.22 | .670 | 17.02 | .670 | 17.02 |
| .032 | 0.81 | .140 | 3.56 | .490 | 12.45 | .696 | 17.68 | .696 | 17.68 |
| .050 | 1.27 | .144 | 3.66 | .493 | 12.52 | .700 | 17.78 | .700 | 17.78 |
| .054 | 1.37 | .146 | 3.71 | .530 | 13.46 | .705 | 17.91 | .705 | 17.91 |
| .068 | 1.73 | .168 | 4.27 | .531 | 13.49 | .715 | 18.16 | .715 | 18.16 |
| .071 | 1.80 | .198 | 5.03 | .535 | 13.59 | .721 | 18.31 | .721 | 18.31 |
| .074 | 1.88 | .204 | 5.18 | .536 | 13.61 | .730 | 18.54 | .730 | 18.54 |
| .076 | 1.93 | .225 | 5.72 | .540 | 13.72 | .743 | 18.87 | .743 | 18.87 |
| .084 | 2.13 | .250 | 6.35 | .555 | 14.10 | .760 | 19.30 | .760 | 19.30 |

FIGURE 1. Dimensions and configurations - Continued.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| .761 | 19.33 | .980 | 24.89 | 1.187 | 30.15 | 1.376 | 34.95 | 1.589 | 40.36 |
| .769 | 19.53 | .984 | 24.99 | 1.192 | 30.28 | 1.377 | 34.98 | 1.595 | 40.51 |
| .772 | 19.61 | .985 | 25.02 | 1.214 | 30.84 | 1.380 | 35.05 | 1.604 | 40.74 |
| .777 | 19.74 | 1.002 | 25.45 | 1.215 | 30.86 | 1.381 | 35.08 | 1.630 | 41.40 |
| .781 | 19.84 | 1.007 | 25.54 | 1.220 | 30.99 | 1.382 | 35.10 | 1.635 | 41.53 |
| .805 | 20.45 | 1.010 | 25.65 | 1.229 | 31.27 | 1.385 | 35.18 | 1.636 | 41.55 |
| .824 | 20.93 | 1.020 | 25.91 | 1.233 | 31.31 | 1.386 | 35.20 | 1.646 | 41.81 |
| .829 | 21.06 | 1.025 | 26.04 | 1.235 | 31.37 | 1.396 | 35.46 | 1.699 | 43.15 |
| .860 | 21.84 | 1.055 | 26.80 | 1.243 | 31.57 | 1.437 | 36.20 | 1.752 | 44.50 |
| .875 | 22.23 | 1.062 | 22.23 | 1.251 | 31.78 | 1.464 | 37.19 | 1.756 | 44.60 |
| .893 | 22.68 | 1.068 | 27.13 | 1.252 | 31.80 | 1.465 | 37.21 | 1.761 | 44.98 |
| .896 | 22.76 | 1.090 | 27.69 | 1.256 | 31.90 | 1.470 | 37.47 | 1.860 | 47.24 |
| .898 | 22.81 | 1.096 | 27.84 | 1.257 | 31.93 | 1.479 | 37.57 | 1.949 | 49.50 |
| .901 | 22.89 | 1.099 | 27.91 | 1.260 | 32.00 | 1.483 | 37.67 | 1.959 | 49.76 |
| .902 | 22.91 | 1.105 | 28.07 | 1.286 | 32.66 | 1.493 | 37.92 | 2.068 | 52.53 |
| .908 | 23.06 | 1.108 | 28.14 | 1.296 | 32.92 | 1.501 | 38.13 | 2.074 | 52.68 |
| .917 | 23.29 | 1.109 | 28.17 | 1.305 | 33.15 | 1.505 | 38.23 | 2.084 | 38.23 |
| .931 | 23.65 | 1.110 | 28.19 | 1.312 | 33.32 | 1.506 | 38.25 | 2.160 | 54.86 |
| .935 | 23.75 | 1.126 | 28.60 | 1.339 | 34.01 | 1.510 | 38.35 | | |
| .947 | 24.05 | 1.127 | 28.63 | 1.340 | 34.03 | 1.511 | 38.38 | | |
| .965 | 24.51 | 1.131 | 28.73 | 1.345 | 34.16 | 1.521 | 38.63 | | |
| .971 | 24.66 | 1.132 | 28.75 | 1.354 | 34.39 | 1.562 | 39.67 | | |
| .974 | 24.74 | 1.135 | 28.83 | 1.358 | 34.49 | 1.574 | 39.98 | | |

(D) NOTES:

1. Dimensions are inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are .005 (0.13 mm) for three place decimals and .01 (0.25 mm) for two place decimals. Angles ±1/2º.
4. Note 5 through 10 applicable to contact only.
5. Dimensions apply after plating.
6. Contact finish: Gold plate .000050 inch min. In accordance with MIL-G-45204. A suitable underplating shall be used. Silver shall not be used. Finish is applied after sealing.
7. 63 all over in accordance with ANSI B46.1-62 unless otherwise specified.
8. Remove all burrs and sharp edges .003 to .005 R.
9. Diameters to be concentric within .004 TIR.
10. Diameters in the same cross-sectional plane to be concentric with each other within .003 TIR.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

This connector mates with MIL-C-83723/75 and MIL-C-83723/77.

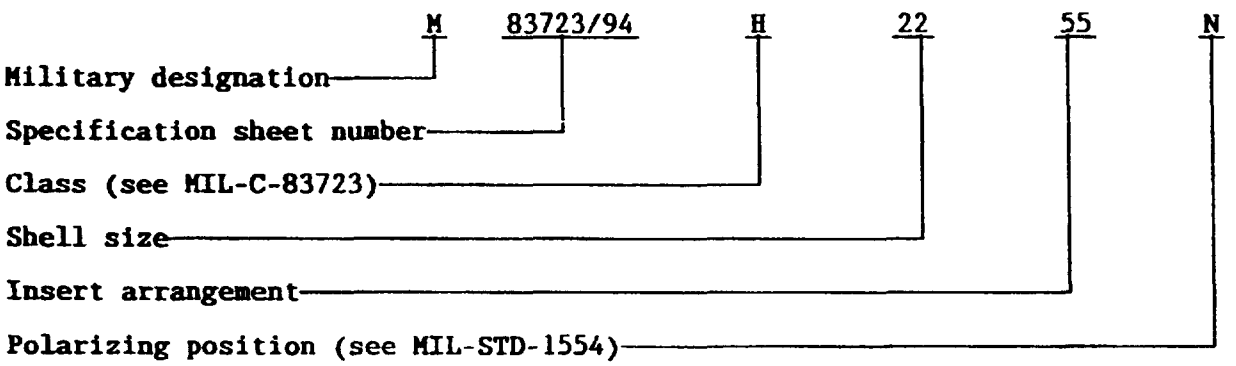
(D) Insert arrangements and polarizing positions: See MIL-STD-1554 and table I.

TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

| Shell size | Insert arrangements | Contact cavities |
|---|---|---|
| 8 | (D) 03 | 1, 2, 3 |
| 10 | (D) 06 | 1, 2, 3, 4, 5, 6 |

(D) Connector accessories: See MIL-C-85049.

(D) Part or Identifying Number (PIN) example:




CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC
Air Force - 11, 15, 99
DLA - ES

Preparing activity
Air Force - 85

Agent:
DLA - ES

(Project 5935-3728-27)



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